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Wheat (*Triticum aestivum* L) is the most extensively grown cereal crop in the world, covering about 220.7 million hectares and accounting for 792 million tonnes. In India, wheat is the second leading cereal after rice and commonly known as Gehu (Hindi), Kanaka (Punjabi), Kotumai (Tamil), Gothambu (Malayalam), Godhuma (Telugu). It is crucial staple food for the Indian population, and plays significant role towards the country's food security and agricultural economy. India achieved the feat of “deficit to surplus” status with the introduction of Mexican dwarf wheat that brought the great Green revolution in India during the sixties. One third of the Indian population depends on wheat as their main food. India is the second largest producer of wheat after China claiming the position of a significant stakeholder in Global wheat production and presently reaping a record production of 113.29 million ton from an area of about 34 million ha with average yield of 3.6 t/ha (IIWBR, 2024). Uttar Pradesh, Punjab, Haryana, Madhya Pradesh, Rajasthan, Bihar, and Gujarat are the major wheat-producing states. Out of all of these states, Uttar Pradesh produces the larger quantity of wheat. The wheat species grown in India are of spring type but cultivated during winter (rabi) season. Among the wheat species, *Triticum aestivum* (Bread wheat), *Triticum durum* (Macaroni wheat) and *Triticum dicoccum* (Emmer or Khapli) are grown in India. In Eastern India bread wheat (*Triticum aestivum*) is generally grown but the productivity is lower than the national level which can be increased by adopting scientific production technologies.



Wheat is a staple ingredient in human diets, providing energy and nutrients like carbohydrates (70%), protein (12%), dietary fibre (2%), fat (1.7%), 2.7% minerals (P, K, Ca, Se, Mn, Zn, Fe and Mg), vitamins (thiamine and vitamin-B), and other bioactive substances with 12% moisture (Aparna *et al.* 2015). It is widely used in making chapati, bread, pasta, biscuits, and other foods, while its straw can be used as fodder, ethanol production, packaging materials and for mulching. Whole wheat, which includes bran and germ, offers medicinal benefits, aiding digestion, and reducing the risks of heart disease, diabetes, and obesity.

Climate requirement:

- ☐ Wheat grows well between latitudes 30°N-60°N and 27°S-40°S.
- ☐ Ideal climate: cool and moist during vegetative growth, followed by dry and warm conditions for grain ripening.
- ☐ Annual rainfall: 750-1000 mm is optimal.
- ☐ Temperature requirements:
 - Germination: 20-25°C
 - Accelerated growth: 20-23°C
 - Grain filling: 23-25°C
 - Ripening: 25-30°C
 - The critical minimum temperature for wheat: 3.5-5.5°C; optimum: 20-25°C; maximum: ~35°C.
 - High (>30°C) or low temperatures during heading, flowering, or maturity cause yield loss.
 - Warm and damp climates or cloudy, humid conditions promote rust diseases.

Soil requirement:

- ☐ Grows well in medium to heavy soils; ideal types are well-drained loamy and clay loam soils.
- ☐ Optimal soil pH: 6-8; tolerable salinity (EC): up to 6 dS/m with moderate amount of water holding capacity.
- ☐ Acidic or highly saline soils are unsuitable; yields reduce by ~50% at 14 dS/m salinity.
- ☐ Soils should have moderate water-holding capacity for better productivity.

Field Preparation:

- Requires well-pulverized, clod-free soil; seeds should be sown at a depth of 5 cm in moist soil for proper germination.
- Conventional tillage involves 2-3 ploughings: one with a soil-turning plough, followed by two to three harrowing or one ploughing with harrow and two ploughing with cultivator followed by planking to retain moisture.
- Pre-sowing irrigation is needed if the soil is dry to ensure proper germination.
- Under reduced tillage system Rotavator-cum-seed drill or Rotavator cum-Fertiseed drill is used in which loosening of soil, levelling, seeding and fertilizer application is done simultaneously.
- If excess moisture is present in the field, then surface seeding of wheat is preferred. Under surface seeding method, seeds are sown 8-10 days prior to rice harvesting. If there is excess moisture in the field after harvesting of rice, then no need to apply irrigation but if the soil is dry, irrigation is to be applied immediately for proper germination of wheat.
- In case of zero tillage, land is not ploughed and sowing of wheat is done at the depth of 2-3 cm with the help of zero tillage machine.

Varieties

Wheat varieties suitable for Eastern UP, Bihar, Jharkhand, Orissa, West Bengal, Assam and plains of NE States are mentioned in table 1.

Table 1: Wheat varieties with their yield and special features

Sl. No	Name of Varieties	Maturity duration (days)	Average Yield (q/ha)	Salient features
Timely sown, Irrigated condition				
1	HD 2733	120-122	48.1	Resistant to brown rust and tolerant to leaf blight
2	HD 2824 (Poorva)	120-122	45.7	Resistant to brown rust and tolerant to leaf blight
3	K0307 (Shatabadi)	120-125	45.6	Tolerant to terminal heat stress
4	Raj 4120	117-124	47.0	Resistant to all three rusts and tolerant to leaf blight
5	HD 2967	125-135	47.0	Resistant to foliar blight and brown rust
6	NW 5054	120-140	47.0	Resistant to foliar blight and brown rust
7	K 1006	120-125	47.0	Tolerance to leaf blight, loose smut & powdery mildew
8	Sabour Samriddhi (BRW 3708)	120-125	46.8	Tolerance to leaf blight and brown rust. Suitable for the state of Bihar
9	DBW 187 (Karan Vandana)	121-123	48.8	Good biscuit spread factor(8.6cm), High Fe content (43.1ppm), resistance to yellow and brown rust
10	HD 3249 (Pusa Wheat 3249)	120-122	48.8	Rich in Iron content (42.5 ppm)
11	DBW 222 (Karan Narendra)	139-150	48.9	Resistant to brown rust, good chapatti quality (7.5/10), bread quality (8.24)
12	PBW 826	145-150	49.7	Resistance to wheat blast
Late sown, Irrigated condition				
1	Narendra Wheat 1014 (NW 1014)	100-130	30.8	High degree of resistance against all the three rusts, leaf blight and karnal bunt
2	HW 2045 (Kaushambi)	100-130	33.7	Tolerant to terminal heat
3	DBW 14	100-105	42.7	Tolerant to leaf blight and terminal heat
4	HD 2985 (Pusa Basant)	105-110	37.7	High level of resistance against leaf blight, foot rot and brown rust
5	Sabour Shreshtha (BRW 934)	105-110	43.1	Resistant to brown rust and loose smut. Suitable for the state of Bihar
6	DBW 107	109-110	41.3	Resistant to brown rust and tolerant to terminal heat

7	HD 3118 (Pusa Vatsala)	109-120	41.7	High degree of resistance to yellow and brown rust
8	DBW 316 (Karan Prema)	113-114	41.0	Resistant to wheat blast and all three rusts, protein content (13.2%), tolerant to drought and heat stress
9	PBW 833	110-115	42.8	Highly resistant to leaf rust, good chapatti quality and protein content (12.9%)
Very late sown, Irrigated condition				
1	HI 1621 (Pusa Wheat 1621)	100-105	28.3	Suitable for very late sown conditions
2	HD 3271 (Pusa Wheat 3271)	100-105	28.1	Suitable for very late sown conditions
Timely sown, Rainfed condition				
1	C 306	130-140	26.0	Good for chapatti quality
2	HD 2888	135-140	22.5	High degree of resistance to brown, black rusts & having high flour recovery
3	Sabour Nirjal (BRW 3723)	129-132	28.7	Resistant to brown rust and loose smut. Suitable for Bihar
4	HD 3171	122 (100-140)	28.01	High degree of resistance to yellow, brown and black rusts, Higher zinc content (47.1 ppm)
5	K 1317	120-128	30.1	Resistant to brown rust and leaf blight. Good Chapatti quality
Late sown, Rainfed condition				
1	K 8962 (Indra)	100-120	20.4	Tolerance to brown and black rust
2	K 9465 (Gomti)	100-120	20.3	Resistant to brown rust and tolerance to heat and drought
Varieties for Sodic soil				
1.	KRL 210,	140 -145	36.1- 56	Suitable for saline and waterlogged areas
2.	KRL 213	140 -145	26.6 - 47.7	
3.	RAJ 3077	120-125	35-40	

Seed treatment:

- ☐ Treat wheat seeds with Vitavax or Thiram at 2.5 g/kg of seed.
- ☐ Use solar or hot-water treatment to control loose smut.

Seed rate and sowing:

- Wheat is cultivated during the winter season, from November to April.
- Optimum sowing time is when mean daily temperature of $23\pm 3^{\circ}\text{C}$.
- Ideal tillering temperature: $16-20^{\circ}\text{C}$.
- Seed rate, spacing and depth of sowing under different conditions are mentioned in Table 2.

Table 2: Sowing time, seed rate and spacing under different condition:

A) Irrigated condition	Seed rate	Spacing	Depth(cm)
a) Timely sowing (First to third week of November)	100 kg/ha	22.5 x 10 cm	5-6
b) Late sowing (Last week of Nov. to first fortnight of December)	125 kg/ha	18 x 10	5-6
c) Very late sowing (mid of December to first week of January)	125 kg/ha	15 x 10	5-6
B) Rainfed condition	100 kg/ha	20 x 10	6-7

Nutrient management

- **Irrigated conditions:** Apply 120:60:40 kg N:P₂O₅:K₂O/ha, full P and K with half dose of N are to be applied as basal, 1/4 N at CRI stage and rest 1/4 N at 40-45 days after sowing.
- **Rainfed conditions:** Apply 90:60:40 kg/ha N:P₂O₅:K₂O as basal at the time of sowing.
- **Zinc management:** Zinc sulphate should be applied @ 25 kg/ha once a year or @10 kg/ha per crop in Zn deficient soils.
- Zinc deficiency can also be corrected by foliar spray of 0.5% zinc sulphate. Prepare the solution for spray by dissolving 2.5 kg zinc sulphate and 1.25 kg un-slaked lime in 500 litres of water. Two to three sprays at 15-day intervals are needed.
- Sulphur deficiency: Apply Cosavet 80WDG @ 8-10 kg/ha at the time of sowing or just before first irrigation.

Integrated nutrient management

- ☐ Apply 75% recommended nitrogen (N) dose + 25% N through organic sources (5-10 t/ha FYM, vermicompost, well decompose poultry manure) + ZnSO₄, @ 25 kg/ha to enhance nutrient availability and maintain soil health.
- ☐ For late-sown crops, use 25% FYM combined with inorganic fertilizers.
- ☐ Apply FYM @ 5 t/ha or other organic matter 5-6 weeks before sowing.

Water management

A wheat crop needs 300 to 400 mm of irrigation water. The most critical stages for irrigation in wheat crop are Crown Root Initiation stage and heading stage. Four to six irrigations are enough for wheat crop. The critical stages of wheat are as follows:

1 st Irrigation	21-25 DAS	Crown root initiation stage (CRI)
2 nd Irrigation	40-45 DAS	Tillering stage
3 rd Irrigation	65-75 DAS	Late joining stage /Boot stage
4 th Irrigation	90-95 DAS	Flowering stage
5 th Irrigation	105-110 DAS	Milking stage
6 th Irrigation	115-120 DAS	Dough stage

Under limited supply of water

- If only one irrigation is possible, apply at CRI stage
- If two irrigations are possible, CRI and Flowering stage
- If three irrigations are possible: CRI, Boot and Milk stage

Weed management

- Wheat fields are typically infested by both monocot and dicot weed species.
- Monocot weeds include *Phalaris minor*, *Avena fatua*, *Polypogon monspeliensis*, *Cynodon dactylon*, *Cyperus rotundus* etc.,
- Dicot weeds include *Chenopodium album*, *Fumaria parviflora*, *Melilotus alba*, *Cirsium arvense*, *Medicago denticulata*, *Sonchus arvensis*, *Polygonum plebejum*, *Alternanthera philoxeroides* etc.
- Weeds can reduce yield up to 40%, so efficient weed management practices are essential to mitigate yield losses.
- Hand weeding twice at 30 and 45 DAS is recommended for better weed control.
- In case of labour shortage, pre-emergence application of Stomp Xtra (Pendimethalin 38.7 % CS) at 600 ml/acre and post emergence application of Total Herbicide (Sulfosulfuron 75% + Metsulfuron 5% WG) at 16 gm/acre during 30 to 35 DAS. The important herbicides for weed management in wheat have been listed in Table 3.

Table 3 : Promising herbicides for weed control in wheat

Herbicides	Dose (g/ha)	Time of application (DAS, days after sowing)	Weed control
Pinoxaden	50	Post emergence (30-35 DAS)	Annual grasses
Clodinafop propargyl	60	Post emergence (30-35 DAS)	Annual grasses, specially <i>Phalaris minor</i> and <i>Avena fatua</i>

Fenoxaprop ethyl	100-120	Post emergence (30-35 DAS)	Annual grasses
Sulfosulfuron	25	Post emergence (30-35 DAS)	Broad-spectrum weed control
Clodinafop propargyl + metsulfuron methyl	60 + 5	Post emergence (30-35 DAS)	Broad-spectrum weed control
Clodinafop propargyl + carfentrazone ethyl	60 + 20	Post emergence (30-35 DAS)	Broad-spectrum weed control
Sulfosulfuron + metsulfuron methyl	25 + 2	Post emergence (25-30 DAS)	Annual grasses and broad-leaved weeds
Mesosulfuron + idosulfuron	12 + 24	Post emergence (20-25 DAS)	Annual grasses and broad-leaved weeds
Isoproturon + metsulfuron methyl	1000 + 4	Post emergence (20-25 DAS)	Annual grasses and broad-leaved weeds
2,4-D	500-750	Post emergence (20-25 DAS)	Annual broad-leaved weeds
Pendimethalin	1000	Pre emergence (0-3 DAS)	Annual grasses and broad-leaved weeds
Flumioxazine	125	Pre emergence (0-3 DAS)	Broad-spectrum weed control
Pyroxasulfone	127.5	Pre- to early post emergence (0-20 DAS)	Broad-spectrum weed control
Flufenacet	300-400	Pre- to early post emergence (0-20 DAS)	Annual grasses

Resource conservation technologies (RCTs) options

- ❑ **Laser land levelling:** Ensures uniform water distribution, reduces water stress or excess in fields, and is essential for adopting other RCTs.
- ❑ **Zero tillage:** Enables wheat sowing by happy seeder or zero-till-seed-cum-ferti-drill machine without field preparation, saving energy, reducing costs, advancing sowing by 4-5 days, requiring less water, and controlling *Phalaris minor*.
- ❑ **Furrow irrigated raised bed system (FIRBS):** Furrow made by furrow raised bed maker. FIRBS saves ~25% seed and fertilizer, and reduces water usage by 25-40% depending on soil and climate.
- ❑ **Rotary tillage:** Pulverizes the top 10 cm soil, places seed and fertilizer simultaneously, and ensures high productivity with low energy requirements.

- ❑ **Turbo happy seeder & rotary disc drill:** Facilitate sowing under loose residue conditions, preventing residue burning, reducing pollution, and conserving soil organic carbon and nutrients.

Cropping system

Rice-wheat, Rice-wheat-green gram, Maize-wheat, Pigeonpea-wheat, Pigeonpea-wheat-green gram, Maize-Toria-wheat, Maize-potato-wheat, Rice-wheat-cowpea, Millets (Jowar/Bajra)-wheat are some of the important cropping systems of eastern India.

Mixed or intercropping of wheat with mustard, gram, linseed are common in most of the wheat growing areas.

Integrated pest management (IPM)

- Foliar blight and brown rust are the main crop health problems of wheat in Eastern India. For effective management of the diseases, cultivation of recommended varieties, like HD 2733, HD 2824, HI 1563, DBW 39, CBW 38, NW 1014, NW 2036, K 9107, HD 2967 (resistant to LB), DBW 14, HD 2888, K0307, DBW39 and HUW 468 should be encouraged.
- The IPM module involves the seed treatment with *Tricoderma viride* (@4g/kg seed) + carboxin (75WP @1.25g/kg seed) or tebuconazole (@ 1.0g/kg seed) for the control of loose smut, followed by broadcast of insecticide treated soil (with endosulfan 35 EC @ 2.3L or chloropyrifos @ 3L/ha) at 15DAS for termites.
- Ear cockle is an important disease in eastern parts of India, hence proper precautions be taken, especially in eastern U.P., Bihar and Jharkhand. The infested seed should be floated in 2 percent brine solution for this purpose The galls will float on the surface. These should be separated and destroyed away from the field by burning. The seed should be thoroughly washed to remove the salt solution before sowing.
- Seed treatment with confidor super (Imidacloprid 30.5% SL) at the rate of 0.3 ml/kg wheat seeds for control of shoot fly infestation and seed treatment with vitavax power (Carboxin 37.5% + Thiram 37.5% DS) at 3g/kg seed for 6 hours is recommended to prevent loose smut of wheat.
- For the management of aphids, foliar spray of imidacloprid 200SL @20g a.i./ha on border rows at the start of the aphid colonization be given. This will help in protection of the bioagent insect, the lady bird beetle inside the field which feeds on aphids.
- For control of Karnal Bunt, the seed crop can be given one spray of propiconazole or two sprays of T.viride at tillering and ear head emergence. For the control of powdery mildew in disease prone areas, one need-based spray of propiconazole (Tilt 25 EC @

0.1%) can be given at ear head emergence or appearance of disease on flag leaf, whichever is earlier.

Harvesting, threshing, winnowing and crop yield

When seeds got dry and turn pale yellow in colour it should be harvested and left in the field to dry. Further, the grains are separated from the ear by threshing which is followed by winnowing and packing. At harvesting stage there should be 25% moisture in the grain. For safe storage grain should be cleaned and dried well in sun for a few days so that the moisture content of the grain comes down to 10-12%. Average yield of wheat varies from 20 to 50 q/ha which depends on the variety, time of sowing and many other factors and yield of some of the varieties are mentioned in Table 2.

References:

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